Adwait V. Datar

Postdoctoral Researcher

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Research profile

Research at the intersection of machine learning, control theory, optimization, and information geometry. I study how the geometry of probability distributions and dynamical systems can inform the design and analysis of learning algorithms. Current questions include natural-gradient optimization, Wasserstein-geometric alternatives to KL divergence, and well-posed regularization for reinforcement learning and optimal control. My doctoral work used robust-control tools, particularly integral quadratic constraints, to study distributed control. More broadly, I aim to connect control theory with machine learning and reinforcement learning.

AI-assisted research

- Routinely use **AI as a research collaborator through OpenAI Codex** for mathematical error analysis, derivation and rigor checks, focused literature review, straightforward calculations, code, and exposition; independently verify mathematical and bibliographic outputs.
- Integrated Codex throughout the Wasserstein and Kalman-Wasserstein KL-divergence project and in ongoing work analyzing errors in the Cole-Hopf transform and extending a Koopman-theoretic result.
- Preparing AI-assisted collaboration workflows for three current bachelor's thesis projects, with emphasis on traceable reasoning and critical validation.

Experience

Postdoctoral Researcher Jul 2023-Present
Institute for Data Science Foundations, TU Hamburg *Hamburg, Germany*

- Develop theory at the interface of information geometry and learning, including natural-gradient optimization and Wasserstein-geometric regularization for learning and control.
- Co-author of work accepted at ICML 2026 and published in *JMLR*, *TMLR*, and *SIAM Journal on Scientific Computing*; teach and mentor across machine learning, information geometry, and control.

Postdoctoral Researcher (6 months) / Doctoral Researcher Nov 2016-Jun 2023
Institute of Control Systems, TU Hamburg *Hamburg, Germany*

- Developed modular methods for distributed control of heterogeneous multi-vehicle systems, using robust-control tools and integral quadratic constraints to analyze uncertain cooperative dynamics; doctorate defended in November 2022.
- Built an indoor multi-quadrotor laboratory and led source-seeking experiments that connected theoretical guarantees with hardware demonstrations.

Research Engineer Jul 2012-Aug 2014
Mercedes-Benz R&D India / Daimler R&D *Pune, India / Stuttgart, Germany*

Education

Dr. rer. nat. (PhD) 2023
TU Hamburg *Hamburg, Germany*
Distributed Control of Multi-Vehicle Systems Using Modular Architectures; defended November 2022.

MSc, Mathematical Modelling in Engineering (MathMods) 2016
University of L'Aquila, University of Hamburg, Gdańsk University of Technology *Europe*
Joint Erasmus Mundus programme; thesis at the University of Stuttgart; grade 1.7.

BE, Mechanical Engineering (with Distinction) 2012
University of Pune *Pune, India*

Teaching and mentoring

- Co-delivered **Machine Learning II** lectures and exercises (winter 2025/26), led **Information Geometry** exercises (summer 2026), and previously taught exercises for Machine Learning I and II.
- Co-developed a course-integrated **Prerequisite Companion** for Machine Learning II: concise notes and diagnostic quizzes designed to close conceptual gaps in a cohort of 30-100 students.
- Organized exercises for **Optimal and Robust Control** (2018-2023), **System Identification** (2018, 2019, 2022), and **Control Systems Theory and Design** (2018); produced reusable and recorded teaching material.
- Supervised **20 bachelor's and master's students**; six projects led to peer-reviewed publications in machine learning, data-driven control, and robotics.
- Delivered an industry tutorial on integral quadratic constraints for IAV GmbH.

Selected research outputs

1. **V. Stein, A. Datar, and N. Ay.** "Well-Posed KL-Regularized Control via Wasserstein and Kalman-Wasserstein KL Divergences." Accepted at *ICML*, 2026. [link]
2. **A. Datar and N. Ay.** "Information Geometry for Wasserstein KL Divergence of Gaussian Measures on $\mathbb{R}^n$." Submitted to *Information Geometry*; arXiv:2503.24022, v3, 2026. [link]
3. **A. Datar and N. Ay.** "Wasserstein KL-Divergence for Gaussian Distributions." *Geometric Science of Information*, 2025, pp. 91-101. [link]
4. **A. Datar and N. Ay.** "Convergence Properties of Natural Gradient Descent for Minimizing KL Divergence." *TMLR*, 2025. [link]
5. **N. Ay, J. van Oostrum, and A. Datar.** "On the Natural Gradient of the Evidence Lower Bound." *JMLR* 26, 2025. [link]
6. **C. Datar, A. Datar, F. Dietrich, and W. Schilders.** "Systematic Construction of Continuous-Time Neural Networks for Linear Dynamical Systems." *SIAM Journal on Scientific Computing* 47(4), 2025. [link]
7. **A. Datar, C. Hespe, and H. Werner.** "Robust Performance Analysis of Cooperative Control Dynamics via Integral Quadratic Constraints." *IEEE Transactions on Automatic Control* 70(1), 2025. [link]
8. **A. Datar, P. Paulsen, and H. Werner.** "Flocking Towards the Source: Indoor Experiments with Quadrotors." *European Control Conference*, 2020, pp. 1638-1643. [link]

Communication and open research

- Presented natural-gradient research at FDIG 2025, University of Tokyo, and at TU Hamburg's Mathematics of Machine Learning 2025 conference.
- Co-created WiMAS, open-source wireless multi-agent simulation software, and produced public course recordings and experimental robotics demonstrations.

Methods and tools

Research: machine learning, information geometry, natural-gradient methods, optimization, reinforcement learning, system identification, robust control, multi-agent systems, and data-driven control.

Tools: Python, MATLAB, LaTeX, OpenAI Codex.

Languages: English, German (B1), Marathi (native).